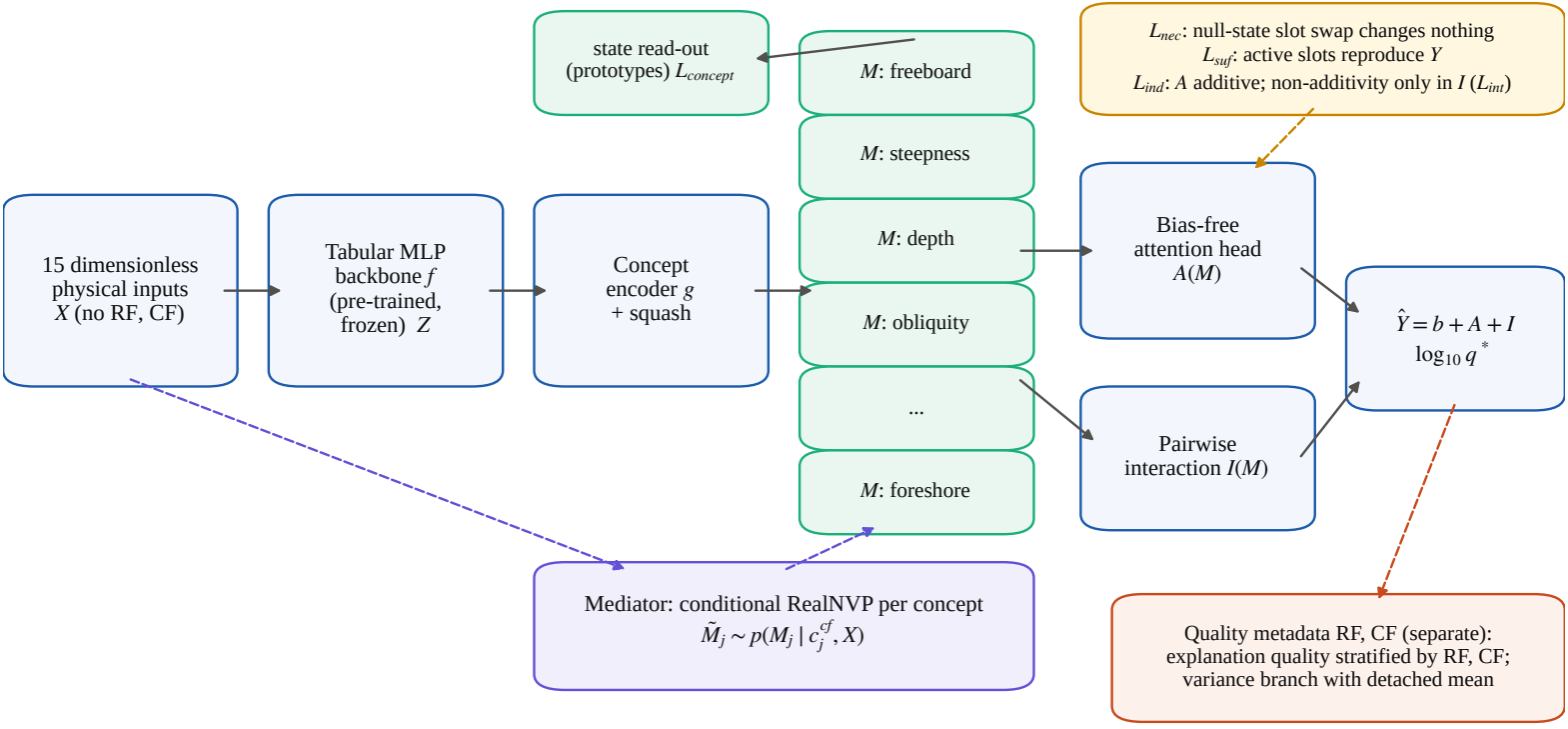
15 dimensionless physical inputs X (no RF, CF)
Tabular MLP backbone f (pre-trained, frozen) Z
Concept encoder g + squash
state read-out (prototypes) $L_{concept}$
M: freeboard
M: steepness
M: depth
M: obliquity
...
M: foreshore
L_{nec}: null-state slot swap changes nothing
L_{suf}: active slots reproduce Y
L_{ind}: A additive; non-additivity only in I (L_{int})
Bias-free attention head $A(M)$
Pairwise interaction $I(M)$
$\hat{Y} = b + A + I$
$\log_{10} q^*$
Mediator: conditional RealNVP per concept
$\tilde{M}_j \sim p(M_j \mid c_j^{cf}, X)$
Quality metadata RF, CF (separate): explanation quality stratified by RF, CF; variance branch with detached mean